

KINETIQ — Hardware Procurement & PCB Specification

Version: 1.0 (Frozen Scope) | Target: ESP32-S3 Wearable | RTOS: Zephyr 4.2+

1. Bill of Materials & Procurement Directory

Item	Locked MPN / Spec	Qty	Source	Footprint / Notes
MCU DevKit	ESP32-S3-DevKitC-1-N16R8	2	Robu.in	DevKit Header Breakout
MCU Module	ESP32-S3-WROOM-1-N16R8	5	Evelta	Castellated SMD Module
IMU (Proto)	BMI270 Breakout	3	Robocraze	Breadboard Breakout
IMU (PCB IC)	Bosch BMI270 IC	5	Robu.in	LGA-14 (2.5x3 mm)
PPG Sensor	MAX30102 Module	3	Robu.in	Module Breakout
Display	GC9A01 1.28" IPS Non-Touch	3	Probots	7-Pin SPI Header
Mic	INMP441 I2S MEMS	4	Robocraze	Breakout / Bottom port
Amp	MAX98357A I2S Class-D	4	Robu.in	Breakout Header
Speaker	8Ω 0.5W Micro 20 mm	4	Robu.in	Wire Leads
Buttons	6x6 mm Tactile SMD	20	Robu.in	4-Pin SMD
Battery	3.7V 600mAh LiPo (WLY602540)	3	Robu.in	JST-PH 2.0 / Leads
Charger IC	TP4056 Linear Charger	5	Robu.in	SOP-8 Package
LDO Regulator	ME6211C33M5G-N 3.3V 500mA	5	Robu.in	SOT-23-5 Package
USB-C Jack	Type-C 16-Pin Receptacle	10	Robu.in	16P SMD + Tabs

2. Component Selection Rationale & Subsystem Integration

- ESP32-S3-WROOM-1-N16R8 (Brain):** Selected for dual-core Xtensa performance, 8MB PSRAM, and native BLE/USB support. Using the integrated-antenna module avoids the severe RF tuning and impedance pitfalls of bare-chip layout on a 2-layer PCB. Mounted at the board edge running Zephyr RTOS tasks.
- ESP32-S3-DevKitC-1 (Bring-Up Target):** Matches the MCU pinout for immediate firmware bring-up while the PCB is fabricated. Serves as a drop-in contingency inside the enclosure if PCB production slips.
- Bosch BMI270 (IMU):** Has in-tree Zephyr driver support ('bosch,bmi270') and onboard step/motion processing. Runs on I2C0 at 100 Hz to generate continuous motion intensity and activity classifications.
- MAX30102 (PPG Sensor):** Integrated optical red/IR sensor supported via Zephyr 'maxim,max30102' compatibility. Faces the wrist on the bottom shell; provides resting HR and quality flags.
- GC9A01 1.28" IPS Display:** Round form factor matches the smartwatch chassis. Controlled over SPI2 via direct-draw framebuffer routines. Eliminating LVGL avoids unnecessary memory overhead and keeps the 5-week timeline on track.
- INMP441 Mic & MAX98357A Amp (Audio):** Digital I2S audio eliminates external codecs and analog noise. INMP441 streams 16 kHz voice over I2S0-RX; MAX98357A drives the 20 mm speaker over I2S-TX without an MCLK line.
- TP4056 & ME6211 Power System:** The ME6211 provides 500 mA output with low ripple for RF and sensors. The TP4056 delivers safe CC/CV LiPo charging via USB-C.

3. PCB Layout & Electrical Design Rules

- 1. ESP32-S3 Antenna Keepout:** Place the WROOM-1 module at the board edge. Maintain a 15 mm keepout boundary with zero copper, ground fills, traces, or vias on any layer under or adjacent to the antenna area.

2. USB-C CC Pull-Down Resistors: CC1 and CC2 must each have an independent 5.1 k Ω pull-down resistor to GND. Never tie CC1 and CC2 together; doing so prevents charging from USB-C PD power supplies.

3. Native USB D+/D- Routing: Route GPIO19 (D-) and GPIO20 (D+) directly to the USB receptacle as a 90 Ω differential pair over an unbroken ground plane. Match trace lengths within 0.5 mm.

4. Shared I2C0 Bus: Terminate SDA and SCL with dedicated 4.7 k Ω pull-up resistors to the 3.3 V rail placed close to the MCU.

5. Passive Sizing: Use standard 0603 footprints for discrete resistors and capacitors to allow reliable hand-assembly and hot-air rework during integration sprints.

6. Power Decoupling: Place a 10 μ F bulk capacitor paired with a 0.1 μ F bypass capacitor directly beside every VDD pin, sensor rail, and display connector.